

Music Similarity Searching via A.I. Transformers

Joy Khera, Randeep Chahal, Lohith Mula
Professor Hao Wang – AI 595 Applied Machine Learning

PROBLEM STATEMENT

- * Streaming services currently lack a feature that recommends similar songs to users based on their music tastes
- * Current song-recommendation algorithms implement collaborative filtering, which utilizes other users’ preferences as opposed to analyzing a song’s internal audio features
- * Collaborative filtering risks recommending songs that significantly differ in style and musical characteristics to a user’s playlist

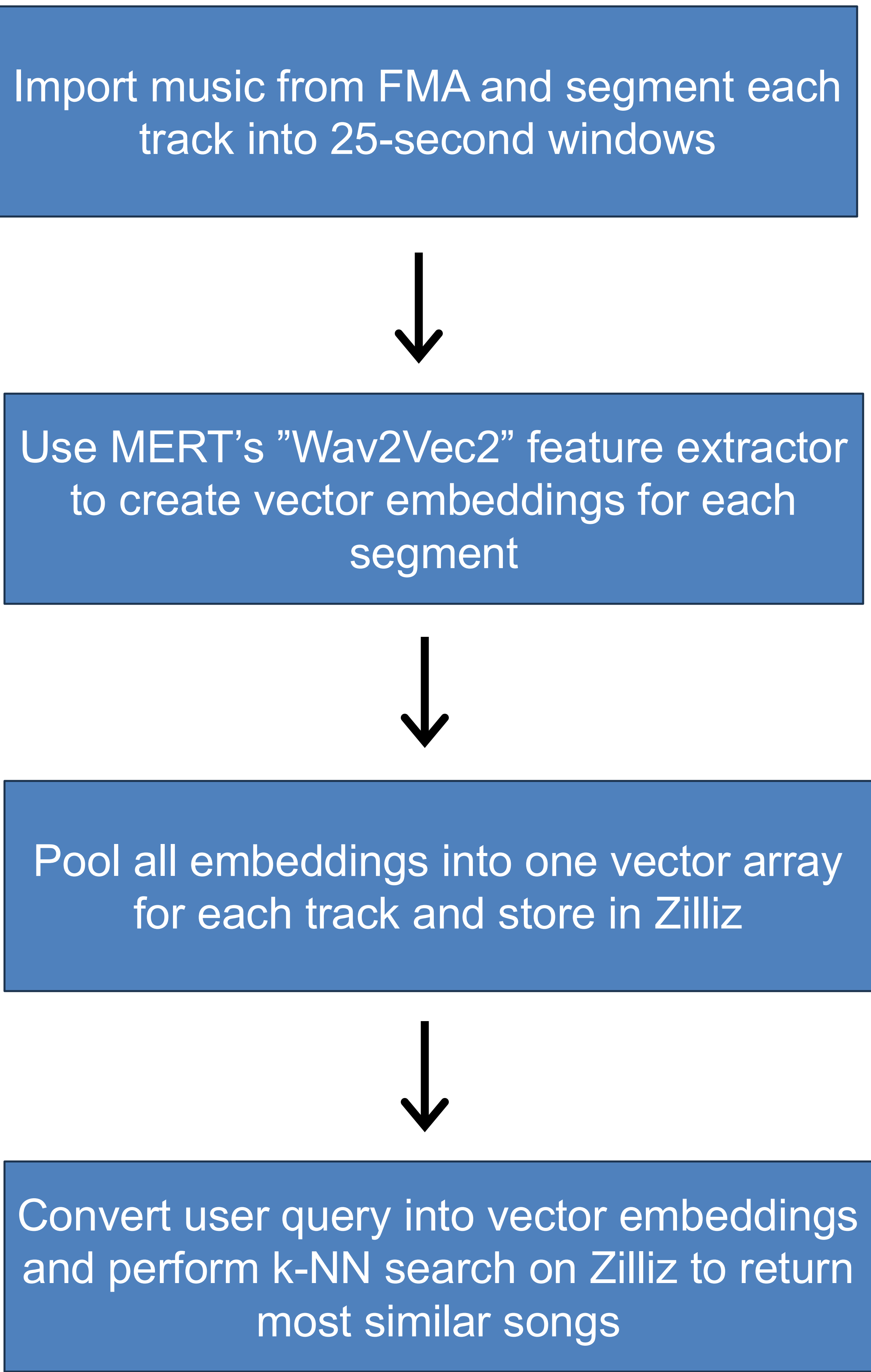
OBJECTIVE

- * Build a music similarity-searching model that recommends similar songs to a user by analyzing an inputted song’s audio features
- * Create an interactive web interface that enables users to input a track and returns songs with the highest similarity percentages

SETUP

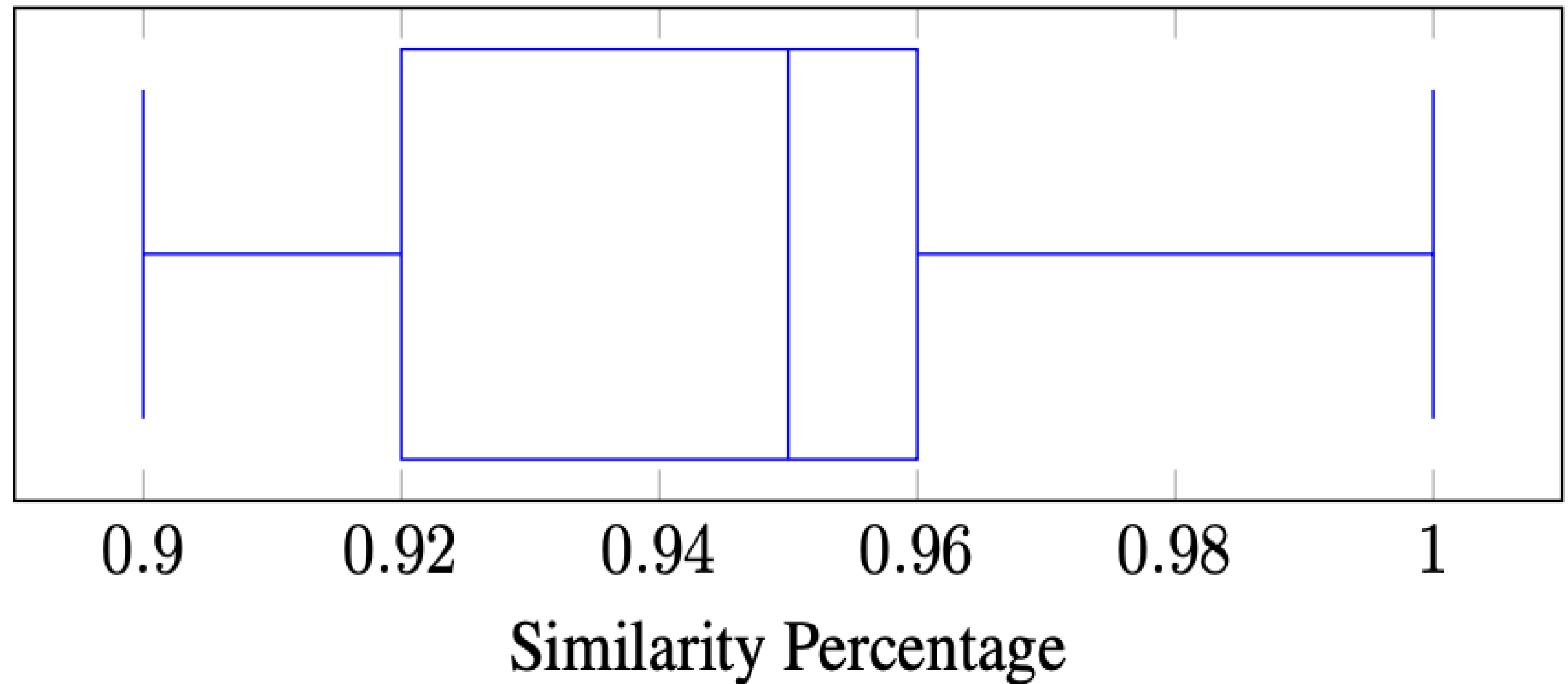
- * **Free Music Archive (FMA):** Publicly available music dataset with ~100k openly licensed tracks
- * **Music Embedding Representation Transformer (MERT):** Pre-trained A.I. transformer model that converts audio tracks into vector embeddings
- * **Zilliz:** Cloud-based vector database

METHODOLOGY



RESULTS

Method	P@1	P@5	P@10
MERT (Ours)	28.00%	29.20%	27.40%
MFCC	25.00%	28.40%	29.10%
Random	12.00%	13.60%	12.40%
Same-Genre	100.00%	100.00%	100.00%



CONCLUSIONS

- * The MERT-based similarity searching model returns songs with a similarity percentage of 90% or greater for its top 10 user-query results
- * MERT has a 25-30% precision rate for its top 10 outputs being of the same genre as the user’s inputted track, indicating an area for improvement